

Aleksandr Razin

TBILISI · GEORGIA

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ML Engineer (~5 years) spanning applied research and product engineering, specialized in **image/video generation & enhancement**. Improved model architectures and post-training optimization from lightweight smartphone to large cloud models, contributing to commercial products. Shipped **LLM/VLM** and multimodal agentic systems to production with API & local inference, serving **2M+ monthly users**. Led a 5-engineer team and worked as an individual contributor across BigTech and startups; **M.Sc. in Machine Learning**.

Experience

Independent Researcher

Nov 2025 – Present

- Authored **Latent Upscaling Adapter** 📄 (accepted **ECCV 2026**) – theoretical research and experiments design

TradingView

SENIOR ML ENGINEER

Sep 2025 – Present

- Engineered the harness and full offline/online evaluation for an agentic **AI copilot** 📄 (hundreds of tools, tens of subagents) for real-time chart analysis & modification; **2M+ monthly users**
- Built production **multimodal** search (text, charts/images, video) on fine-tuned embeddings + hybrid retrieval; **~80% precision** per modality
- Cut existing LLM-solution costs **~20%/month** by blending third-party APIs with local inference

Startup

FOUNDER, LEAD ML ENGINEER

Apr 2025 – Sep 2025

- Founded and hired a **5-engineer** product team; secured **~€100K**; **acquired (2025)** by a large IT company
- Shipped a retail **video-analytics** CV suite (detection, segmentation & tracking) to 3 of top-5 national retailers, saving **~€20K+/month**
- Post-trained **text-to-image / image-to-image** ad-generation models, improving identity preservation and lifting client ad **CTR ~15%**

Huawei Research Institute

ARTIFICIAL INTELLIGENCE RESEARCHER

Dec 2023 – Apr 2025

- Redesigned **video restoration** architectures (attention + convolution optimization), improving perceptual quality, detail reconstruction & color
- Optimized **diffusion training** (checkpoint-merging & loss design) for high-frequency texture synthesis; **~10%** gains over baselines (FID/PSNR)
- Led **quantized post-training** for the **Mate 70** 📄 night-HDR video restoration network (enhancement, super-resolution, joint denoising & demosaicing); refined strategy cut ghosting & flicker, shipping weekly to production

RESEARCH INTERN

May 2023 – Sep 2023

- Improved domain **LLM** generation **> 30% ROUGE** via classification-agent query routing and LoRA + DPO post-training
- Enabled **multimodal** LLM agents by training metric-learned visual encoders on 100K+ sample datasets for visual question answering

ITMO University | Industry AI Laboratory

DEEP LEARNING ENGINEER

Jun 2022 – May 2023

- Built an end-to-end **data pipeline** (extraction, labeling, processing, analysis), raising detection & segmentation Precision/Recall
- Distilled banknote segmentation models (**20% fewer FLOPs**, no accuracy loss); **contrastive learning** lifted fake-stamp detection +10% F1

Mining University | Robotic Laboratory

SOFTWARE ENGINEER

Jan 2021 – Dec 2021

- Built **on-device** software and trained lightweight models (detection, classification, segmentation) for obstacle & defect detection on **pipeline robots** 📄 at oil & gas plants, cutting manual inspection by **~€56K+/year**

Education

ITMO University

M.SC. MACHINE LEARNING, GPA: 4.9/5 – THESIS: *Enhancing Image Super-Resolution through Depth Estimation*

2022 – 2024

Teaching Assistant: *Advanced Machine Learning Technologies*

Mining University

B.SC. ENGINEERING, GPA: 4.9/5 – THESIS: *Real-time obstacle detection and analysis for robotic exploration with LiDAR*

2018 – 2022

Teaching Assistant: *Experimental Physics*

Mathematics Tutor (Volunteer): *Olympiad mathematics & competitive programming*

Skills

Languages Python, Go, JAX (basic)

Tools Docker, Kubernetes, Kafka, AWS, Git, CI/CD, CUDA Toolkit, MLflow, Airflow, Kubeflow

Frameworks PyTorch, TorchVision, OpenCV, Transformers, Diffusers, DeepSpeed, DiffSynth-Studio, ComfyUI, vLLM, Google ADK

Concepts Vision / Language / Multimodal models, distributed training, model & training optimization, post-training, experiment management, A/B testing, MLOps, SWE

Language Skills English (fluent), Russian (native)

Selected publications

2026	A. Razin , D. Kazantsev, I. Makarov. <i>One Small Step in Latent, One Giant Leap for Pixels: Fast Latent Upscale Adapter for Your Diffusion Models</i> . Hugging Face Daily Paper #1 	<i>ECCV 2026</i>
2024	Gorbatosvski, A. V., Razin, A. D. , Aliev, A. A., Kovalchuk, S. V. <i>Improving question answering in programming domain with pretrained language model fine-tuning using structured diverse online forum data</i> . Journal of ITMO 	<i>Journal</i>
2022	Pshenin, V., Liagova, A., Razin, A. , Skorobogatov, A., Komarovsky, M. <i>Robot crawler for surveying pipelines and metal structures of complex spatial configuration</i> . MDPI Infrastructures 	<i>Journal</i>

Awards

ELLIS PhD Program: shortlisted & offered a PhD with Jinjin Gu (ESRGAN) & Radu Timofte (Swin Transformer) on video generation & world models (top 15% of 6,000+ applicants)	<i>2026</i>
Huawei Internship: after successful participation, offered a full-time position as a CV Engineer	<i>2023</i>
Kaggle: “OTTO – Multi-Objective Recommender System”: top 10% out of 2,574 teams 	<i>2023</i>
Grant winner, Russian Student Startup: top 5% out of ~10k teams (~€15k funding)	<i>2023</i>
Prize holder, Russian Olympiad in Phys. & Math. “Star”: Spacecraft track, top 2% of ~8k participants	<i>2018</i>
School Gold medal: excellent academic performance at high school	<i>2018</i>